

What you get on 27 October

A **works-level dataset over a frozen frame of all 4,299,418 Canadian works**, in which every record carries: **why it was retrieved** (route provenance), **which sense of “Canadian” it satisfies**, language, **its category** (metaresearch, metaepidemiology, and the traditions the call names, each defined from its own literature), the **loannidis domain** for metaresearch works, **study design, data and code links**, and a **confidence**. With it: the **cited inclusion criteria and the census they were derived from**; a **preregistered bilingual human audit** giving prevalence, retrieval sensitivity and **differential error by language, source and abstract availability**; every prompt, label and disagreement; and a **bilingual explorer, already live at metacan.xera.ac**.

The frame is **built, not sampled**: all **482 partitions** of a pinned OpenAlex snapshot, each work exactly once. The pilot below is done, public, and re-derivable offline with no network.

Don’t search for metaresearch. Search for Canada, then screen.

Retrieve what *looks* like metaresearch and the field’s boundary becomes a property of your keyword list; the map is then unauditable, because you cannot measure what a lexicon never showed you. I inverted it: the frame is **all Canadian research**, an external checkable criterion, and membership becomes a **classification** over that frame rather than a **retrieval** over the literature. Retrieval fails both ways and we measured both. The best topic route **finds 12% of the field**, because OpenAlex files a work by what it is *about*, so metaresearch about cardiology reads as cardiology: **the problem is not vocabulary, it is aboutness**. And it wrongly *admits*, with **17,466 works** entering on a Canadian “institution” that does not exist. **1,565,226 works (36.4%) carry no Canadian affiliation at all**, so an affiliation-only frame never sees them.

The criteria, derived rather than asserted

Three frontier models screened the same **5,600 works**, drawn with known selection probabilities, against one locked rubric, on identical prompts, with labels written by the harness and reconciled by set equality.

Rate agreement is not set agreement. Of the **274 works** any model called metaresearch, only **104 (38%)** were called metaresearch by **all three**; **117 (43%)** rest on a single model’s opinion; pairwise overlap is about **50%**. At a ~1% base rate the settled rejects buy near-total agreement for free. **The boundary is not a line the models share; it is a region they each cut differently**, stable at $n = 1,000, 2,000$ and $5,600$.

The field has no consensus definition either, and says so in print. Puljak et al. analysed **175 published sources**, found “*definitions varied*”, and concluded the “*research community would benefit from consensus*” (J Comp Eff Res 2020); Kataoka et al. **publicly dispute them** (J Clin Epidemiol 2023;154:219); Stevens & Laynor report there is **no MeSH heading** for meta-research (J Med Libr Assoc 2023). **Those are claims about humans. Mine are claims about models. I do not conflate them**: the literature says the field lacks consensus; my pilot says three models cut one rubric differently. Consistent, not identical, and **only the human audit closes the gap**.

So I asked *why* they split, and made it measurable. An independent judge (**a fourth model, not one of the three arms**, and **blind**: opinions shown as A/B/C in random order, no model names) adjudicated **all 179 contested works** and had to name, for each, **which seam of the rubric produced the split**. It shows **no arm preference** ($p = 0.49$). Its verdict: **the rubric was silent on 159 of the 179 (89%)**; only **17 splits** were a screener misapplying a rule that existed.

That turns adjudication into a **frequency-weighted census of the sentences the rubric was missing**. I published the **census, wrote the revision against it, and locked the revision before re-screening**, so it could not be tuned until the numbers improved. **Re-screened by the same three models: 96 of 179 (54%) are now unanimous; pairwise overlap moves 0.19 to 0.36**. Those works were *selected for disagreement*, so this shows the rules resolve the cases they were written for; it is **not** a frame-wide claim, and the unbiased test, a full re-screen, is prespecified. **This is the call’s “inclusion and exclusion criteria to define the community”, derived rather than asserted**, with the residual **83 works** released as the ones all three models still split on.

What broke, and what now runs. My sampling design **could not reach 12.9% of the frame**. $\text{sum}(N_h) = N$ is the one identity a stratified design rests on and I never checked it, so **549,370 works had inclusion probability exactly zero**, including **328,912 Canadian-affiliated works about Canada**, which is **the secondary estimand’s own cell**. Nothing failed; every stratum returned exactly the n it asked for. It was found by a model I asked to attack the design. Seven strata now **partition** the frame by construction, asserted on every build. **DEVIATIONS.md carries all twenty-six entries; every one produced output that looked correct, and none threw**.

1. Methodology

No category is defined from memory. All seven are locked and cited, each quoted verbatim from a source committed to the repository: *metaresearch* (Ioannidis, PLoS Biol 2015;13:e1002264, with his five thematic areas as a per-record domain); *bibliometrics* (Mingers & Leydesdorff 2015); *sts* (Jasanoff 2004; Latour); *scholarly_communication* (Borgman 2007); *open_science* (UNESCO 2021); *research_integrity* (Fanelli 2009; COPE). *metaepidemiology* is coded under **both** published definitions (Murad 2017; Kataoka 2023) and **flagged where they disagree**, because a live dispute between named researchers is a boundary to be *measured*, not settled by fiat. **The screen is multi-label**: a bibliometric study of citation

distortion is *both*, and forcing the choice was the bug. “**Adjacent**” is defined by negation, and a category meaning only *near the important one* cannot be applied consistently, which is exactly why **OUT-vs-adjacent was our largest confusion at every sample size**. The models were not failing; **the category was empty**. It was also a judgment I had no standing to make: STS has its own founding literature and is not an appendage of metascience. The build **fails** if any category the schema allows is undefined in the rubric.

Estimand, prespecified. *Primary:* Canadian-produced metaresearch (≥ 1 Canadian affiliation **or** funder). *Secondary:* metaresearch **about** the Canadian research system. **71.2% of the frame carries no funder metadata**, so both clauses rest on sparse metadata: the frame unions four routes, every record carries **provenance**, and the audit samples what none reached.

Screening stratifies; it never measures. Machine labels ship *provisional*. Design-weighted estimates are unbiased however noisy the stratifier is **only while every record keeps a nonzero selection probability**, a condition **my own design violated for 12.9% of the frame** before it was repaired and made a build check. The classifier therefore **decides where humans look and never supplies a label**; its **4.8×** lift is lift toward *machine* labels, and is not validated efficiency until the humans code.

Annotation, gated on measurement. The LLM never touches 4.3M works: it labels **~10,000**, a classifier trains on those, and inference over the frame is cheap. **The risk is not cost, it is validity**, and we measured it once already: a classifier distilled from our *metaresearch* labels matched **its own teacher at Jaccard 0.17**. **Shipping that across 4.3M works would publish 4.3M confident errors**. So **study design ships only where it can be checked**. MEDLINE publication types are **human-indexed by NLM (31,442 Canadian works tagged *Randomized Controlled Trial*)**, so design labels are **validated against NLM’s indexers on held-out data** and released only for classes clearing a prespecified threshold. **MEDLINE is biomedical**: outside it no such standard exists, so those works carry a **score, not a label**, marked unvalidated. **The dataset states, per field, which of its own annotations it has earned the right to assert**.

Validation, because machine screening cannot establish accuracy and sampling a 3.2M rejected mass for its misses returns an expected 0.4 of them. **(i) A stratified probability sample with guaranteed floors** (machine score, **between-model disagreement**, language, source, abstract availability), with a baseline draw from **every** stratum so every record keeps a **known nonzero** selection probability; two humans code independently, blind to machine labels. **(ii) Known-item recall on external criteria**: venue, registries, and MEDLINE publication types, all immune to the aboutness that defeats topic retrieval.

Sample: $n \approx 1,000$, dual-coded. **The second coder is the single largest delivery risk and is not optional**: I recruit and pay a bilingual (FR/EN) coder in Week 1 via CaRN, ACFAS, CAIS-ACSI and CARL; **if none is found the French claim is withdrawn, not fudged**. **Feasibility, after costing it wrong twice (D7):** the stratifier reads every work against the full rubric (**\$1,094**) plus a 20,000-record second pass (**\$15**). **Total \$1,110, leaving ~\$1,790 for the coder, who is the study**.

2. Openness, transparency, reproducibility

The repository is **public** and the committee can run it today. *make pilot-offline* re-derives **every figure above** from archived responses **with no network**; *make proposal* **fails the build if this page quotes a number the pilot does not produce** (it caught three while this page was being written, including a round number I had invented); *make lint* fails if the strata do not partition the frame or if the codebook contradicts its schema. **Each guard exists because that failure already happened**. Extraction is from a **pinned S3 snapshot**, not the metered API (8.2 days per pass).

One reproducibility claim I will not make: temperature 0 does not make an LLM deterministic and these versions will be retired, so I release the **exact prompts and raw outputs** but do **not** claim the labels are reproducible. Each release carries a Zenodo DOI: dataset (Frictionless, CC-BY-4.0), code (MIT), Docker, lockfiles, **every rubric version, labels with their disagreements**, the **seam census**, validation results *including the errors*, a **Datasheet**, **DEVIATIONS.md**. Preregistered on **OSF**.

3. Diversity, assumptions, limitations, bias

I am a solo applicant working in English. I do **not** claim to represent the field’s diverse traditions; that needs collaborators I do not have, and I once claimed to have *measured* how badly I serve them and **withdrew it** (four records, $p = 0.141$). What I can do is stop defining anyone by their distance from me: **each tradition is defined from its own literature, and cited**.

Érudit, the main francophone Canadian platform, matches 0 OpenAlex sources (OAI-PMH live, 379 sets): the francophone literature is not *thin* here, it is largely **absent**. French power comes from the **Érudit harvest**, verified but not yet run. Funder metadata skews against the traditions the call asks me to include (**NSERC carries 9.2×** **SSHRC’s linked works**), and **31.5% of the frame has no abstract**, where the screen finds half as much metaresearch ($p = 0.023$).

Assumptions named. The base rate rests on machine labels: **a hypothesis with a denominator, not a result**. Coverage holds *within the frame*, not over all Canadian metaresearch. **Governance:** no sensitive identity is inferred; where Indigenous-governed data are implicated no derived labels are released; OCAP® without an Indigenous partner is compliance theatre. I submit as an independent researcher with no affiliation: **one of the people this dataset cannot see**.

4. Work plan (1 Aug to 27 Oct)

W1–2 Preregister (OSF); **recruit and pay the bilingual coder**. · **W3–5** Freeze the Érudit harvest; screen the full frame; **report the version-to-version difference**. · **W6–8** Stratified sample from screened-in **and** screened-out; dual blinded coding; adjudication. · **W9–10** Design-weighted estimation; the differential bias report; annotation validated against MEDLINE. · **W11–12** Release; deploy; plenary.